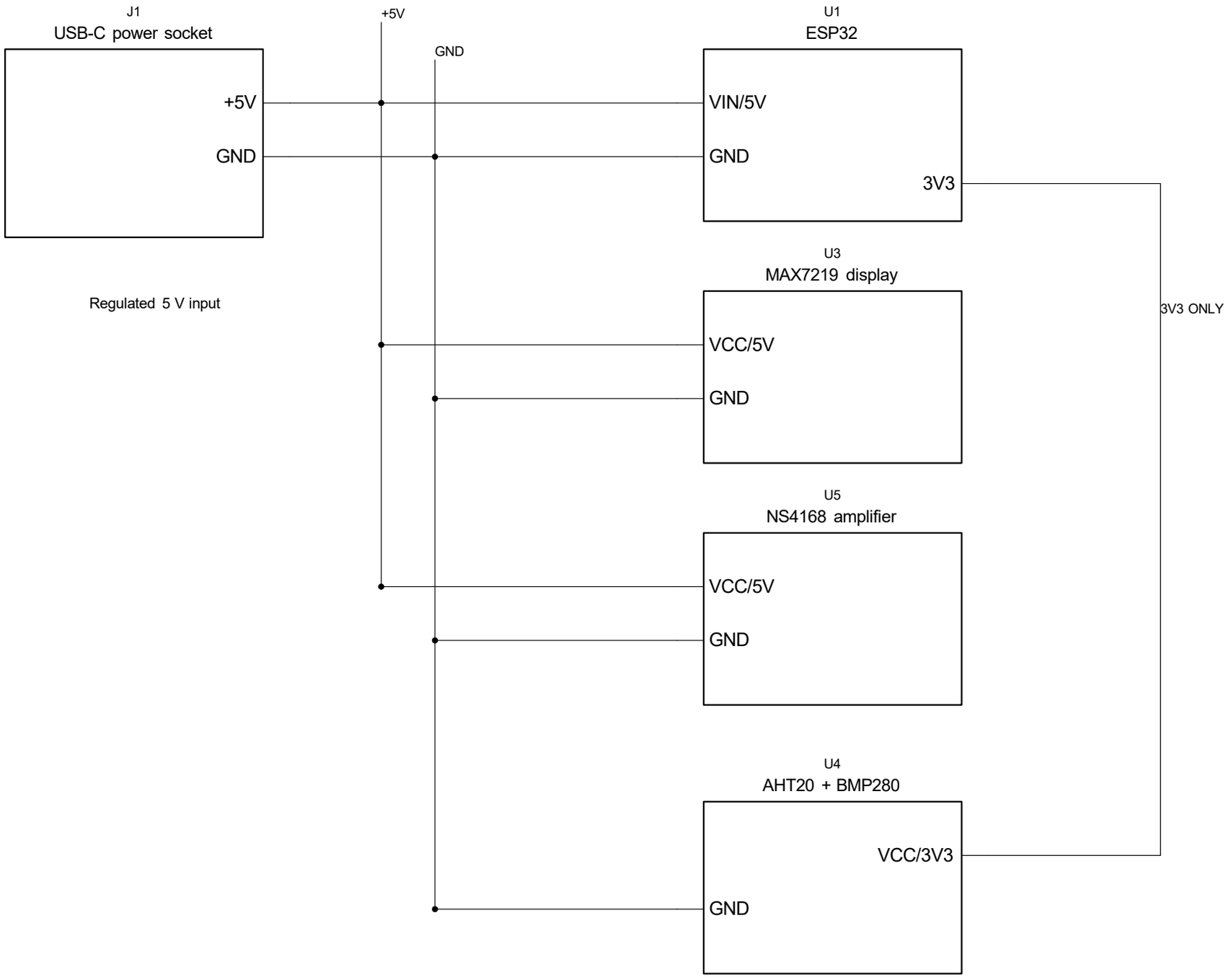
USB-C POWER | DIRECT SUPPLY DISTRIBUTION

All 5 V loads connect in parallel to J1. ESP32 3V3 powers only the sensor.

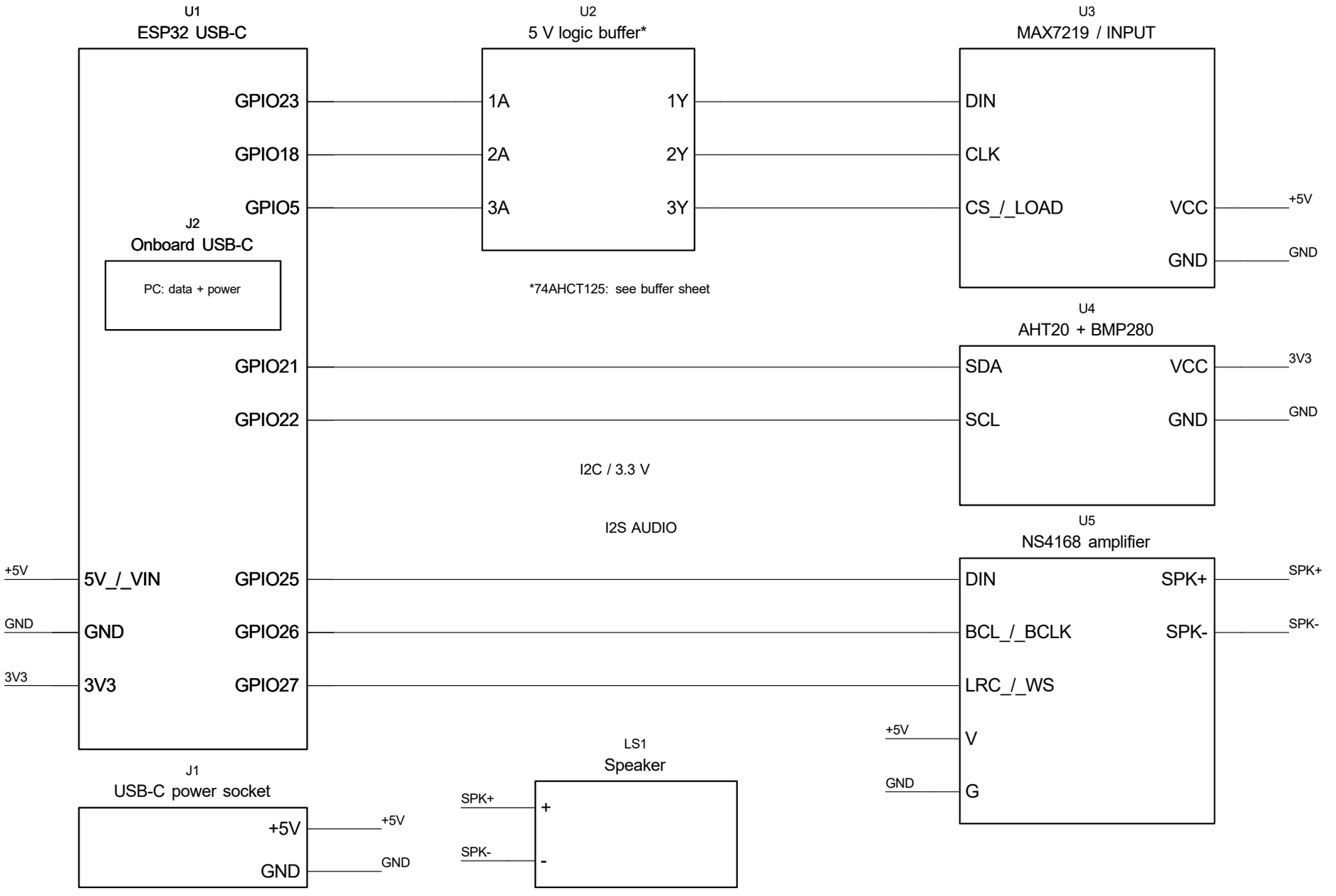


Dots mark connected branches. Crossings without dots are NOT connected.

Never join +5V and 3V3. Disconnect external VIN before PC USB power.

ESP32 SMART CLOCK | MODULE WIRING

Connect by printed terminal labels. GPIO numbers are not header positions.



SPK- is an amplifier output. NEVER connect it to GND.

J1: external regulated 5 V supply. J2: ESP32 programming USB-C.

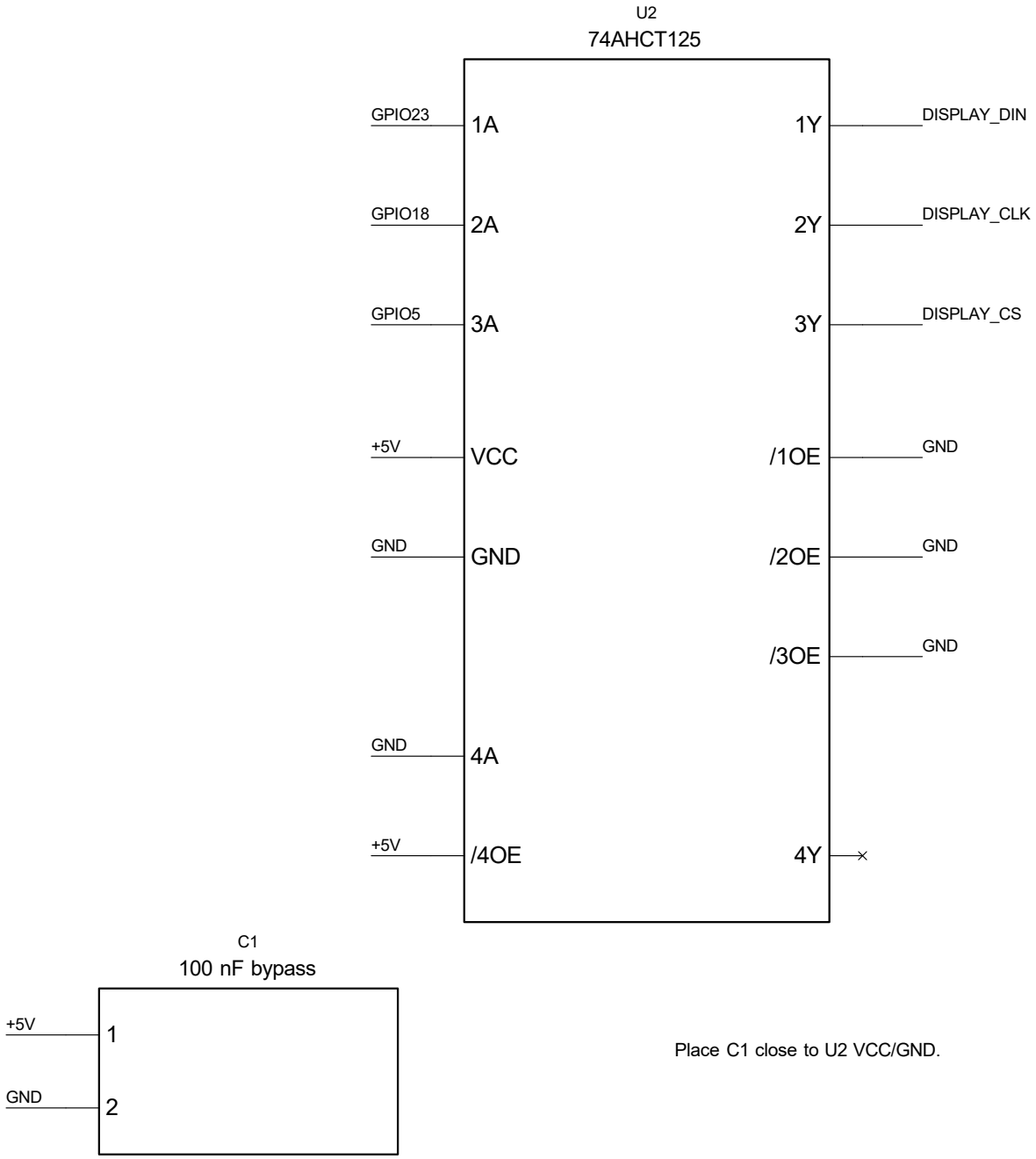
All identical net labels connect together. Sensor VCC uses 3V3 only.

Disconnect external VIN power before powering ESP32 from PC USB.

Module-level schematic | firmware: esphome-smart-clock.yaml | NirBY

DISPLAY BUFFER | 74AHCT125

Recommended for guaranteed 3.3 V to 5 V logic levels



Place C1 close to U2 VCC/GND.

Terminal names shown; use your IC package datasheet for pin numbers.

/OE means active LOW. Tie unused input 4A LOW and /4OE HIGH.

The direct-wired prototype does not include this extra buffer.